



POLICY FORUM

Independence by permission

Industry-academy collaboration explores the 2020 US election

By **Michael W. Wagner**

In the spring of 2020, pressed by internal research staff and with support from the highest levels of the company, Facebook (now Meta) began a collaboration with outside academics to study the 2020 US presidential election. Interest in social media's role in elections came after evidence that the 2016 US election was rife with suspicious and divisive groups targeting swing-state residents on Facebook (1). As one of the initiatives the team used to demonstrate transparency, I was approached to serve as the project's independent rapporteur. I conclude that the team conducted rigorous, carefully checked, transparent, ethical, and path-breaking studies—reported by

González-Bailón *et al.* (2) and Guess *et al.* (3, 4) on pages 392, 398, and 404 of this issue, respectively, and by Nyhan *et al.* (5)—in the first public results of this project. Though the work is trustworthy, I argue that the project is not a model for future industry-academy collaborations. The collaboration resulted in independent research, but it was independence by permission from Meta.

The research used platform-based interventions of consented participants to show that both altering what individuals saw in their newsfeeds on Facebook and Instagram and altering whether individuals encountered reshared content affects what people saw on the platforms. However, these changes did not reduce polarization or improve political knowledge during the 2020

US election. Indeed, removing reshared content reduced political knowledge.

After 3 years, the outcomes of these experiments are the first publicly shared results from this US 2020 Election Project (hereafter, “2020EP”). The project is also producing papers on topics such as whether Facebook enables ideological segregation, whether campaign advertising on platform influences attitudes and behaviors, the effects of coordinated inauthentic behavior (i.e., disinformation spreading), and a comprehensive description of Meta's political communication ecosystem. Several papers are working their way through the peer review process.

The 2020EP is a partnership between Meta and 17 scholars working at universities across the US (hereafter, “outside academ-

ics”) (6). More than two dozen researchers, engineers, legal analysts, and others worked on the project for Meta. The research leads were Chad Kiewiet de Jonge, Annie Franco, and Winter Mason (7). The principal investigators of the study were outside academics Natalie (Talia) Jomini Stroud and Joshua Tucker. All of the 17 outside academics on the team had a prior connection with Social Science One, a project that had previously piloted some research partnerships, with at best mixed results, between academic researchers and Facebook.

The team hoped my role would increase the confidence that the public and the scholarly community would have in the project. My charge includes commenting upon the nature of the collaboration between Meta and the outside academics and whether the project might serve as a model for future industry-academy collaborations. I was not compensated by Meta or the outside academic team. I knew some members of both teams before I began my work. Since June, 2020, I have observed more than 350 virtual research meetings and 2 days of in-person research meetings, totaling more than 500 hours. I conducted 41 interviews with Meta researchers and staff and members of the outside academic team. I also interviewed two former Meta employees, five major social science research funders, and five academic experts in the area who were not a part of the project. I have access to drafts of working papers and appendices, many of the conversations between the research team that took place over the web app Workplace, project code, drafts of project-governing documents, meeting agendas, some email correspondence, and observations of team members at conferences and in-person work sessions.

As I elaborate below, my conclusion is that for social science research about the effects of social media platforms and their use to be truly independent, outside academics must not be wholly reliant on the collaborating industry for study funding, must have access to the raw data that animates analyses, must be able to learn from internal platform sources about how the platforms operate, and must be able to guide the prioritization of workflow. Additionally, some project structures that were appropriate to US-based faculty are unlikely to apply to other parts of the world.

THE BIRTH OF THE PROJECT

Research about how industry and the academy collaborate tends to focus on issues

of research productivity, patents acquired, economic value, future collaborations, and conflict-of-interest concerns. Behaviors most associated with successful collaborations include flexibility with institutional factors, honesty in relationships, clarity in outputs, and sharing an awareness of the frameworks of the collaboration (8). In their model for industry-academy partnerships, King and Persily suggest Social Science One’s structure of outside funding, platform-provided data, and an academic council (9). However, Social Science One faced intense criticism for major delays from Facebook in data sharing, (the company cited privacy concerns) and for a major error in the data found long after Facebook provided it (10, 11). One funder in the social media space that I interviewed called Social Science One a “rope a dope,” arguing that Meta strings researchers along with a promise for data that never comes, or comes with untenable compromises. Despite the middling track record of the Social Science One partnership, Meta researchers argued that though there were multiple potential pathways to pursue the 2020EP, selecting scholars with connections to Social Science One was efficient and expedient. Scholars outside of the 2020EP project, including blind reviewers of articles in this issue, have criticized the project for its US-centric focus and for only including those with Social Science One connections, leaving out less well-resourced scholars.

Meta researchers were keen to conduct rigorous research of the 2020 election with outside academics to add a critical layer of legitimacy to their work. Several Meta employees said that it was personally and professionally important to them to “course correct” after the emotional contagion study and the Cambridge Analytica scandals revealed that millions of unconsented users had their data collected to manipulate their emotions or expose them to political advertising, respectively (12). Meta researchers wanted a study that asked important scientific questions, engaged in rigorous analysis, respected platform users’ privacy, practiced consented research, and collaborated with respected scholars. In addition to the internal advocacy at Meta, Meta faced pressure from the academy, the public, and lawmakers to share their knowledge, resources, and data.

In early 2020, Meta approached Tucker and Stroud to serve as co-principal investigators (PIs) for the project. The co-PIs selected the rest of the academic team without Meta’s input or approval, though they agreed to constrain themselves to the nearly 100 North American-based scholars affiliated with Social Science One.

GUARDRAILS

The outside academics and Meta researchers generally agreed about what would make the project a success. They wanted to conduct rigorous, transparent, and ethical studies about important questions related to social media’s effects on the 2020 US presidential election. Consistent with research highlighting the importance of agreeing upon outputs, the team agreed to study four key outcomes across more than a dozen papers: political participation, political polarization, knowledge and misperceptions, and trust in US democratic institutions. The team agreed that each study should be preregistered, specifying hypotheses, research designs, measurement strategies, how effect sizes would be interpreted, and reserving the right to conduct additional analyses.

Additionally, the team felt it was important that the studies be credible—and be perceived to be credible. Thus, the outside academics would forego compensation from Meta. The outside academics asked for, and received, “control rights” for the papers. This meant that in the event of a dispute between the outside academics and the Meta researchers over how to interpret a finding (or if there was a dispute between outside academics co-authoring the same paper) or what to include in a research paper, the lead author—which would always be an outside academic—would have the final decision. The only caveat to this rule would be if Meta’s legal team determined there was a privacy issue or other legal constraint that might prevent publication. To date, that has not occurred.

The outside academic team’s PIs indicated on several occasions that if Meta refused to publish the findings of any study, they would walk away from the project. To address the potential that Meta’s leadership would learn of results prior to publication and seek to either curtail publication or create a pre-emptive public relations strategy, Meta researchers and outside academics agreed to not share any results outside of their research team until the papers had been through the peer review process. To my knowledge, this rule was violated once, at a private dinner in which a member of the outside academic team discussed the results of a paper with others. This led to a series of reiterations of the rule with the full team. Additional “guardrails” for the project included:

- Privacy requirements would allow only Meta employees to handle raw data. Outside academics were permitted and encouraged to develop and check code, but could not have the data on their computers.
- As much of the data as possible would be made available to the broader academic community for future scholarship on the

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Inter-university Consortium for Political and Social Research (ICPSR) (13).

- Participants whose individual-level data would be analyzed would provide informed consent prior to participating. This included more than 75,000 participants in experimental analyses and 400,000 participants who took part in at least one of the six survey waves, from August 2020 to March 2021.
- All coauthors would be credited.

Although many people not connected to the project praised the guardrails, others argued that some of the choices made by the team—such as taking no payment from Meta—would prevent less well-resourced US academics and academics from many other countries from pursuing a col-

laboration with Meta, potentially limiting the kinds of research questions asked and methods used to answer them.

and privacy issues within Meta. The leads created a Roles & Responsibilities document (which was completed 2 years into the project) and developed a variety of new procedures on an ad hoc basis to address unforeseen circumstances. As members of both the Meta and outside academic teams regularly said, “we are building the plane while we are flying it.”

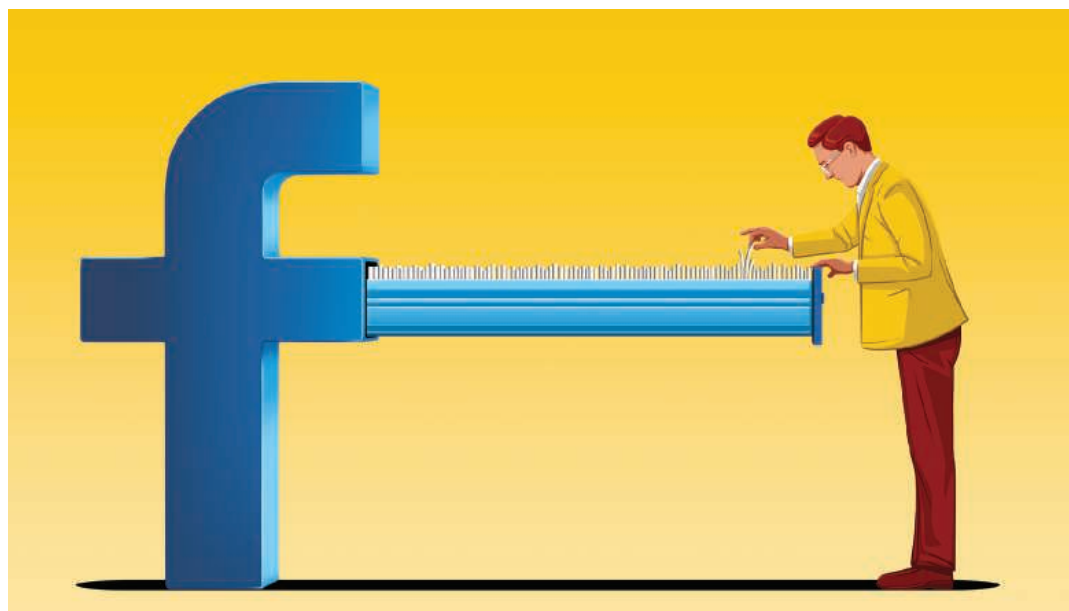
One contribution of the project is the ongoing set of procedures the research team created, followed, and altered over time. A massive quality assurance initiative was launched in real time to check for errors. Rules relating to research designs, as well as data transparency and analysis, were developed in an on-going fashion for the first 2 years of the project. The research team felt the tension between a

researchers could share findings from the papers not yet published in limited settings before those papers completed any peer review because the project has taken so long.

In general, the Meta team agreed to pursue the topics and research designs that outside academics wanted to pursue. The nature of the collaboration allowed the outside academic researchers to peer inside Meta in a way that would have been unavailable to them without the industry collaboration. On occasion, this, and advice they received from Meta researchers, led the outside academics to reshape what it is they sought to study, or how they sought to study it, as they learned details about what data were collected and structured by Meta for analysis. For example, the outside

academics were not precisely aware of how Facebook groups were joined and, in some cases, left, by users. Coming to understand this led to changes in the working paper examining behavioral polarization. Additionally, the academics learned about corporate cultural behaviors at Meta, on topics ranging from paid time off (Meta staff did not work, unlike many of the outside academics who regularly worked while on vacation) to the privacy reviews from Meta’s legal team that were necessary to pass before the papers could be published. Meta researchers also noted that some outside academics were not aware of what

could be analyzed on platform. Although this process provided clear advantages for outside academics as they could benefit from institutional knowledge, one social media expert who was not on the team pointedly noted that, “what data is made available shapes what is asked and answered,” an issue for readers to actively consider when, for example, interpreting the degree to which the results reflect independent user behavior as compared to Facebook and Instagram’s algorithms and other platform features affecting user behavior. A former Meta employee warned, “Facebook researchers will answer every question they get from the professors honestly; they are ethical professionals. However, they are also corporate employees. If the professors don’t ask the exact right question, Facebook staff won’t volunteer that they know how to get what you (out-



laboration with Meta, potentially limiting the kinds of research questions asked and methods used to answer them.

OBSERVATION OF COLLABORATION

My observations of research team meetings for each of the individual papers and weekly planning meetings between the lead Meta researchers and the lead outside academic researchers revealed a deep commitment to high-level social science scholarship. Conversations were substantive, detailed, generally collegial, and usually focused on scientific questions of research design, measurement, coding, data analysis, internal Meta policies, Meta’s data architectures, and quality control.

Meta project lead and outside academic PI conversations also focused on how the work would be received by reporters, other academics, and the public, as well as legal

preference to codify their roles and responsibilities in advance and the considerable time pressure to design and preregister the studies before the election season. In some ways, this process hewed to best practices in industry-academy collaborations with respect to the clarity of outcomes. In other ways, it highlighted the importance of flexibility in industry-academy collaborations as many processes changed substantially over time.

For example, the guardrail of not discussing a paper’s findings until it had been through peer review has changed considerably. Early in the project, the guardrail meant “accepted for publication.” Some team members suggested that “peer reviewed” could be redefined as being achieved even if a paper was rejected for publication. In the spring of 2023, the team began moving toward a policy that

side academics) are really asking for.”

After learning of the Social Science One data error in the fall of 2021 (10), the 2020EP team began discussing and developing major quality-control protocols, checking data and code with multiple sets of eyes across the team. This process took months to complete and persists to this day on some working papers. Although researchers across both groups are confident that the data still contain errors, as all datasets do, more was done to mitigate consequential errors on this project than on any social science research project that any of them, or I, could name.

Although Meta researchers always consulted and sought agreement with the outside academic PIs on project workflow choices, they structured the choices to prioritize the experimental analyses. This was generally efficient; Meta researchers created (and repeatedly revised) a Gantt chart to describe the proposed timeline of progress across the papers, informing the outside academics that it was faster to analyze the experiments as compared to the data gathering and analysis required for the other, nonexperimental papers. However, some Meta researchers felt that the experimental analyses were less likely to produce large effect sizes on the outcomes of interest. Though speculative, this could mean that this first wave of published articles will get the most public attention—because they are the first papers released in the project—even though other papers may produce results that show larger substantive effects.

It is impossible to study everything in one project, and this project’s enormous scope played a considerable role in the length of time it is taking to complete papers. However, two major topics that did not receive much attention were race in the 2020 elections and the ways in which Facebook and Instagram follower networks affected various outcomes of interest. Some of the outside academic team members noted that the very nature of the team, which did not include scholars primarily known for studying race, explained the omission. Others argued that the issue was one of statistical power, whereas others noted that Meta does not collect racial identifiers of its users. Regarding studying the follower networks of the study participants to obtain a more comprehensive understanding of social media’s role in the election, Meta said no, citing privacy concerns. One outside academic said, “it was the only time I heard a hard no, and I understood—privacy—but if you want to understand how social media works, you need to study who follows who.”

A SUCCESS, BUT NOT A MODEL

Perhaps the biggest problem with the 2020EP is that it worked so well; the high quality of the scholarship may encourage interpretations that the project is a model for future social media company–academy collaborations. Researchers were permitted to study almost everything they asked to study. The outside academics had the final say regarding how findings were interpreted and presented in the papers; and though I have observed no reason to believe that this is likely to change, it is the case that as these articles approached publication, Meta sought to reopen policies and procedures related to control rights. Specifically, Meta researchers wanted to be able to expressly disagree with lead author interpretations of findings or other matters in articles they co-authored. Stroud and Tucker opposed this effort, noting that collaborators can remove their name from a paper if they have a fundamental disagreement.

The study contained extraordinary data and code checking processes—a system that caught errors and slowed down the project by several months. The study was conducted ethically. There was informed consent approved by multiple universities and an independent institutional review board (IRB), albeit the IRB from the same company that conducted the survey research elements of the project (NORC; formerly known as the National Opinion Research Center); and the original outside IRB contacted to review the project declined to review it for what I believe are public relations reasons, as compared to research ethics concerns. There was rigorous peer review after detailed preregistration. The studies produced new knowledge that was shared with the world without edit from Meta’s legal or corporate teams. Meta demonstrated a strong commitment to rigorous social scientific research about the ways in which Facebook and Instagram influence elections in the US, spending more than \$20 million and redirecting the time of dozens of employees to serve the project. Concerns that Meta would leak negative study results have not materialized to date.

But Meta set the agenda in ways that affected the overall independence of the researchers. Meta collaborated over workflow choices with the outside academics, but framed these choices in ways that drove which studies you are reading about in this issue of *Science*. Moreover, the collaboration has taken several years and countless hours of time—limiting the ability of the outside academics to pursue other research projects that may have shaped important public and policy conversations.

One shortcoming of industry-academy collaboration research models more generally, which are reflected in these studies, is that they do not deeply engage with how complicated the data architecture and programming code are at corporations such as Meta. Simply put, researchers don’t know what they don’t know, and the incentives are not clear for industry partners to reveal everything they know about their platforms.

In an era in which state and national lawmakers proffer a general rhetorical and increasingly policy-related antipathy toward higher education, independent scholarship is more important than ever. In the end, independence by permission is not independent at all. Rather, it is a sign of things to come in the academy: incredible data and research opportunities offered to a select few researchers at the expense of true independence. Scholarship is not wholly independent when the data are held by for-profit corporations, nor is it independent when those same corporations can limit the nature of what it studied. Creative social media–academy-funder partnerships, or, more likely, government regulation and data-sharing requirements (e.g., the European Union’s Digital Services Act) that also provide privacy protections, as well as defined structures to encourage and protect industry-employed researchers to collaborate, are necessary to foster opportunities for path-breaking, comprehensive scholarship that does not require a social media platform’s permission. ■

REFERENCES AND NOTES

1. Y. M. Kim et al., *Polit. Commun.* **35**, 515 (2018).
2. S. González-Bailón et al., *Science* **381**, 392 (2023).
3. A. M. Guess et al., *Science* **381**, 398 (2023).
4. A. M. Guess, *Science* **381**, 404 (2023).
5. B. Nyhan et al., *Nature* **10.1038/s41586-023-06297-w** (2023).
6. Meta, “Research partnership to understand Facebook and Instagram’s role in the U.S. 2020 election”; <https://research.facebook.com/2020-election-research/>.
7. T. Stroud, J. A. Tucker, A. Franco, C. P. Kiewit de Jonge, “A proposal for understanding social media’s impact on elections: Rigorous, peer-reviewed scientific research,” 31 August 2020; https://medium.com/@2020_election_research_project/a-proposal-for-understanding-social-medias-impact-on-elections-4ca5b7aae10.
8. R. Rybníček, R. Königsguber, *J. Bus. Econ.* **89**, 221 (2019).
9. G. King, N. Persily, *PS Polit. Sci. Polit.* **53**, 703 (2020).
10. C. Timberg, “Facebook made big mistake in data it provided to researchers, undermining academic work,” *Washington Post*, 10 September 2021; <https://www.washingtonpost.com/technology/2021/09/10/facebook-error-data-social-scientists/>.
11. J. Mervis, *Science*, 24 September 2019; <https://www.science.org/content/article/privacy-concerns-could-derail-unprecedented-plan-use-facebook-data-study-elections>.
12. E. Selinger, W. Hartzog, *Res. Ethics Rev.* **12**, 35 (2016).
13. Press Release, “ICPSR to host data on social media’s impact on US 2020 elections”; <https://www.icpsr.umich.edu/web/about/cms/4535>.

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